

Lemma F for the octahedron: opposite petals never overlap

working notes

September 2026 — draft

Audit correction, 9 September 2026. The unrestricted double-outward half-plane claim in the previous draft is false; an integer-coordinate counterexample is checked by rational intervals in `n6/audit_witness.py`. Lemma 5 below is now restricted to the angle range used in Case A. Case B needs a separate treatment of larger angles; its historical reductions below are conditional on that restriction, not a complete partition of all configurations. The hinge reduction also needs the local nonoverlap hypothesis. Neither these notes nor the numerical experiments prove the full six-vertex theorem. See `n6/REVIEW.md` for the full audit.

Setting and conventions

We use the conventions of `HANDOFF.md`. P is a convex polytope of octahedron type with antipodal (non-adjacent) vertices v, w and equator $u_0, \dots, u_3$ (the neighbours of w , indices mod 4, in counter-clockwise order around w seen from outside). The faces are $W_i = wu_iu_{i+1}$ and $V_i = vu_iu_{i+1}$; κ_x is the curvature at x ; ω_i, ν_i are the angles of W_i at w and of V_i at v . The net Z_k cuts $\text{star}(v) \cup \{wu_k\}$: the fan $W_k, W_{k+1}, W_{k+2}, W'_{k-1}$ around w (angular gap κ_w at the slit) with the petal V_i glued to W_i along u_iu_{i+1} . Touching counts as non-overlapping.

Only nine face pairs of Z_k share no net vertex: the three local pairs at the slit, the two opposite-petal pairs $(V_k, V_{k+2}), (V_{k+1}, V'_{k-1})$, and four petal-vs-far-fan pairs. Lemma F is the statement that under the hypothesis

$$\textbf{(H)} \quad \kappa_v \geq \kappa_x \quad \text{for every vertex } x \text{ of } P \quad (1)$$

none of the six far pairs overlaps, in every Z_k . (The handoff's hypothesis $\kappa_v \geq \max_i \kappa_{u_i}$ is weaker; the numerics below show that the weaker hypothesis is *not* enough for the argument of §3, although no counterexample to the weaker statement itself is known.)

Elementary consequences of (H). Since $\sum_x \kappa_x = 4\pi$ over six vertices, (H) gives $\kappa_v \geq 2\pi/3$, hence $\sum_i \nu_i = 2\pi - \kappa_v \leq 4\pi/3$. At any convex vertex each face angle is at most the sum of the other face angles (the spherical polygon inequality for the tangent cone), so $\nu_j \leq \frac{1}{2} \sum_i \nu_i$, i.e.

$$\nu_j \leq \pi - \frac{1}{2}\kappa_v \leq \frac{2\pi}{3} \quad \text{for every } j. \quad (2)$$

1 Reduction to three consecutive petals

Lemma 1 (petals do not cross interior hinges). *Assume the three local pairs at the slit do not overlap. In Z_k no petal properly crosses the relative interior of a hinge segment wu_j , $j \neq k$.*

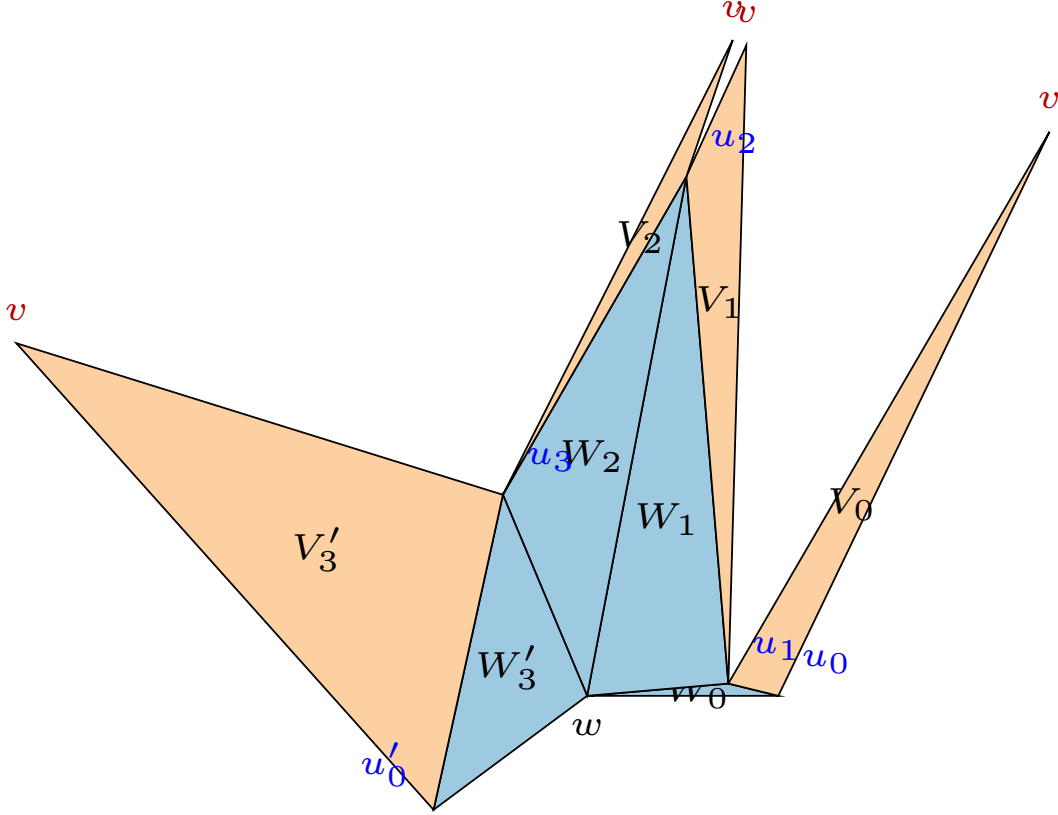


Figure 1: A typical net Z_k (fan faces blue, petals orange), v the sharpest vertex, slit at the sharpest neighbour u_0 of w .

Proof. The two fan faces adjacent to wu_j are W_{j-1} and W_j ; a petal crossing the segment would enter both. Away from the slit, a petal shares a net vertex with W_{j-1} or W_j , and the corresponding wedges are disjoint. There are two exceptions to this shared-copy argument: V'_{k-1} at wu_{k+1} and V_k at wu_{k-1} . In these cases a crossing would enter a local fan face, excluded by the hypothesis. $\square$

Lemma 2 (petal versus far fan face). *Assume the local pairs do not overlap. If a petal V_i overlaps a fan face W_j with $j \notin \{i-1, i, i+1\}$ then V_i overlaps V_j .*

Proof. V_i contains no vertex of W_j (w and the equator vertices of W_j lie in fan faces or petals that share a vertex with V_i , except possibly the local face W'_{k-1} , whose treatment is part of Lemma L). By Lemma 1 the boundary of V_i cannot enter W_j through its hinges, so it enters through the outer edge $u_j u_{j+1}$, which is the common edge of W_j and V_j ; near that crossing V_i meets V_j . $\square$

Subject to Lemma L, Lemma F follows from the following statement about three consecutive petals, applied to the triples (V_k, V_{k+1}, V_{k+2}) and $(V_{k+1}, V_{k+2}, V'_{k-1})$ of Z_k .

Conjecture 3 (Triple Lemma). *Let W_i, W_j, W_{j+1} ($j = i+1$) be three consecutive fan faces of a net and V_i, V_j, V_{j+1} their petals. Under (H), V_i and V_{j+1} do not overlap.*

Numerical evidence 4. No overlap of opposite petals under (H) in 72,000 random nets, nor under the weaker hypothesis $\kappa_v \geq \max \kappa_u$; adversarial hill-climbing (edges ≥ 0.05 diam, curvatures ≥ 0.05) brings the two petals to within 0.4% of the diameter but never into overlap. All observed overlaps (44 in 72,000 nets, without hypothesis) have two equator vertices sharper than v .

2 Where a meeting can be

Fix a triple as in Conjecture 3 and write $u = u_j$, $u' = u_{j+1}$ for the base of V_j , ϕ, ψ for the base angles of V_j at u, u' , $\kappa = \kappa_u + \kappa_{u'}$. In the net, V_i and V_j meet at u only, separated there by the angular gap κ_u between the two copies uv_j and uv_i of the cut edge uv ; likewise at u' . Let L be the line through uv_i and L' the line through $u'v_{j+1}$.

Lemma 5 (double-outward wedge, restricted angle range). *Assume $0 < \phi + \kappa_u < \pi$ and $0 < \psi + \kappa_{u'} < \pi$. V_i lies in the closed half-plane of L not containing V_j , and V_{j+1} in the closed half-plane of L' not containing V_j . Hence every common point of V_i and V_{j+1} lies in $\mathcal{D} :=$ the intersection of these two half-planes, which is the wedge at the point $p^* = L \cap L'$ opposite to the base uu' . If $\kappa < \nu_j$ then $\mathcal{D}$ lies on v_j 's side of the base line (above), and if $\kappa > \nu_j$ it lies on w 's side (below); $\mathcal{D} = \emptyset$ if $\kappa = \nu_j$.*

Proof. The rays uv_i and $u'v_{j+1}$ make angles $\phi + \kappa_u$ and $\psi + \kappa_{u'}$ with the base, so they converge above the base iff $\phi + \psi + \kappa < \pi$, i.e. $\kappa < \nu_j$; the rest is the description of the four wedges at p^* . $\square$

Remark 6. The range assumptions are automatic in Case A, since their sum is $\pi - \nu_j + \kappa < \pi$. They do not follow from (H) in general. Without them even the assertion that V_j is on the opposite side of L from V_i can fail. The rational counterexample in `n6/results/` refutes this unrestricted assertion, not the Case-A proposition.

Lemma 7 (the two cut edges never cross). *In Case A ($\kappa < \nu_j$), the segments uv_i and $u'v_{j+1}$ are disjoint.*

Proof. If they crossed at p then p would be above the base with base angles larger than ϕ, ψ , so v_j lies inside the triangle $uu'p$ and $|up| + |u'p| > |uv_j| + |u'v_j| = |uv_i| + |u'v_{j+1}|$, contradicting $|up| \leq |uv_i|$, $|u'p| \leq |u'v_{j+1}|$. $\square$

Claim 8 (observed overlaps). *In the 44 observed opposite-petal overlaps (no hypothesis), the intersection region lies in $\mathcal{D}$; 26 are above the base and 18 below, exactly as predicted by the sign of $\nu_j - \kappa$. In every below-base case the outer edges uu_i and $u'u_{j+2}$ of W_i, W_{j+1} converge on w 's side (equivalently $\angle_u W_i + \angle_u W_j + \angle_{u'} W_j + \angle_{u'} W_{j+1} < \pi$) and exactly one of the two fan faces extends past the crossing point; both cannot, since fan faces are disjoint.*

3 The rotation picture (above-base case)

Glue V_i, V_j, V_{j+1} flat around v with no gaps (the “ v -fan”); this is possible because $\nu_i + \nu_j + \nu_{j+1} < 2\pi$, and in it the three petals occupy disjoint wedges at v . The net position of V_i is obtained from its v -fan position by the rotation ρ_u about u by the gap angle κ_u (away from V_j), and that of V_{j+1} by the rotation $\rho_{u'}$ about u' by $\kappa_{u'}$. Both rotations have the same sense. Therefore

$$V_i \cap V_{j+1} \neq \emptyset \text{ in the net} \iff \mathcal{R}(V_i^{\text{fan}}) \cap V_{j+1}^{\text{fan}} \neq \emptyset, \quad \mathcal{R} := \rho_{u'}^{-1} \circ \rho_u,$$

and $\mathcal{R}$ is the rotation by $\kappa = \kappa_u + \kappa_{u'}$ about the point c^* on w 's side of the base with $\angle c^*uu' = \kappa_u/2$, $\angle uu'c^* = \kappa_{u'}/2$ (composition of two rotations). Let M be the line c^*v ; $\mathcal{R}$ moves points from u' 's side of M towards u 's side. Measure angles around c^* from M , positively towards u 's side.

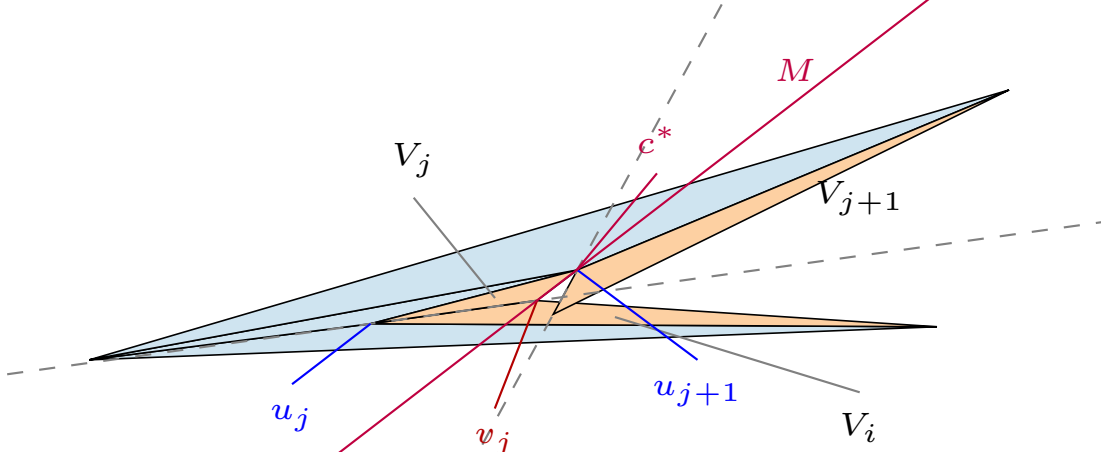


Figure 2: A thin above-base overlap (no hypothesis): the dashed cut-edge lines, the point c^* and the line M of §3. Here $\nu_i + \nu_j > \pi$ and V_i crosses M .

Lemma 9 (angular criterion). *Let $\varepsilon_i \geq 0$ be the largest angle (seen from c^*) by which $V_i^{\text{fan}} \cap \{\text{above the base}\}$ crosses M towards u' 's side, and ε_{j+1} the corresponding excursion of $V_{j+1}^{\text{fan}} \cap \{\text{above}\}$ towards u 's side. If $\kappa < \nu_j$ and*

$$\varepsilon_i + \varepsilon_{j+1} < \kappa, \quad (3)$$

then V_i and V_{j+1} do not overlap in the net.

Proof. By Lemma 5 a common point p is above the base. The rotation ρ_u^{-1} maps the wedge of V_i at u in the net, $[\phi + \kappa_u, 2\pi - \angle_u W_j - \angle_u W_i]$, onto the v -fan wedge, so $x = \rho_u^{-1}(p)$ is below the base only if p is; hence $x \in V_i^{\text{fan}}$ and $y = \rho_{u'}(p) \in V_{j+1}^{\text{fan}}$ are above the base and $y = \mathcal{R}(x)$. Their angles from M satisfy $\theta_y = \theta_x + \kappa$ with $\theta_x \geq -\varepsilon_i$ and $\theta_y \leq \varepsilon_{j+1}$, so $\kappa \leq \varepsilon_i + \varepsilon_{j+1}$ (no wrap-around since $\kappa < \nu_j < \pi$). $\square$

In particular, if neither petal crosses M above the base ($\varepsilon_i = \varepsilon_{j+1} = 0$), there is no overlap. When does a petal cross M ? Write $\mu_u = \angle uv c^*$, $\mu_{u'} = \angle c^* v u'$, so $\mu_u + \mu_{u'} = \nu_j$ (c^* lies in the cone of V_j at v , being on w 's side of the base between the lines vu , vu'). The wedge of V_i at v starts at vu and turns away from V_j by ν_i ; it reaches the ray of M beyond v iff $\nu_i > \pi - \mu_u$. The other way for $V_i^{\text{fan}} \cap \{\text{above}\}$ to cross M is through the far vertex u_i or through the point where the edge vu_i or uu_i crosses the base line.

Lemma 10. *Under (H) the two petals cannot both cross M through their wedges at v .*

Proof. That would need $\nu_i + \mu_u > \pi$ and $\nu_{j+1} + \mu_{u'} > \pi$, hence $\nu_i + \nu_{j+1} + \nu_j > 2\pi$, whereas $\nu_i + \nu_j + \nu_{j+1} \leq \sum \nu \leq 4\pi/3$. $\square$

Lemma 11 (the above-base case needs a very sharp v). *Under (H), $\kappa < \nu_j$ forces $\kappa_v > 6\pi/7$, hence $\sum_i \nu_i < 8\pi/7$ and each $\nu_i < 4\pi/7$.*

Proof. $4\pi = \sum_x \kappa_x \leq 4\kappa_v + \kappa$ and $\kappa < \nu_j \leq \pi - \kappa_v/2$ by (2). $\square$

Numerical evidence 12. In 3000 random octahedra (all apexes, all triples with $\kappa < \nu_j$): the composition $\mathcal{R}$ has the stated centre and sense, and the criterion of Lemma 9 reproduces the net overlap test exactly. Under (H) neither petal ever crosses M above the base ($\varepsilon_i = \varepsilon_{j+1} = 0$ in all 224 triples, degenerate ones included), and adversarial search under (H) cannot make either excursion

positive. Under the weaker hypothesis $\kappa_v \geq \max \kappa_u$ a petal can cross M and (3) can fail (excursion 1.96 with $\kappa = 0.05$), so (H) is genuinely used. Every one of the 26 observed above-base overlaps violates $\varepsilon_i = \varepsilon_{j+1} = 0$ (20 through V_i , 6 through V_{j+1}).

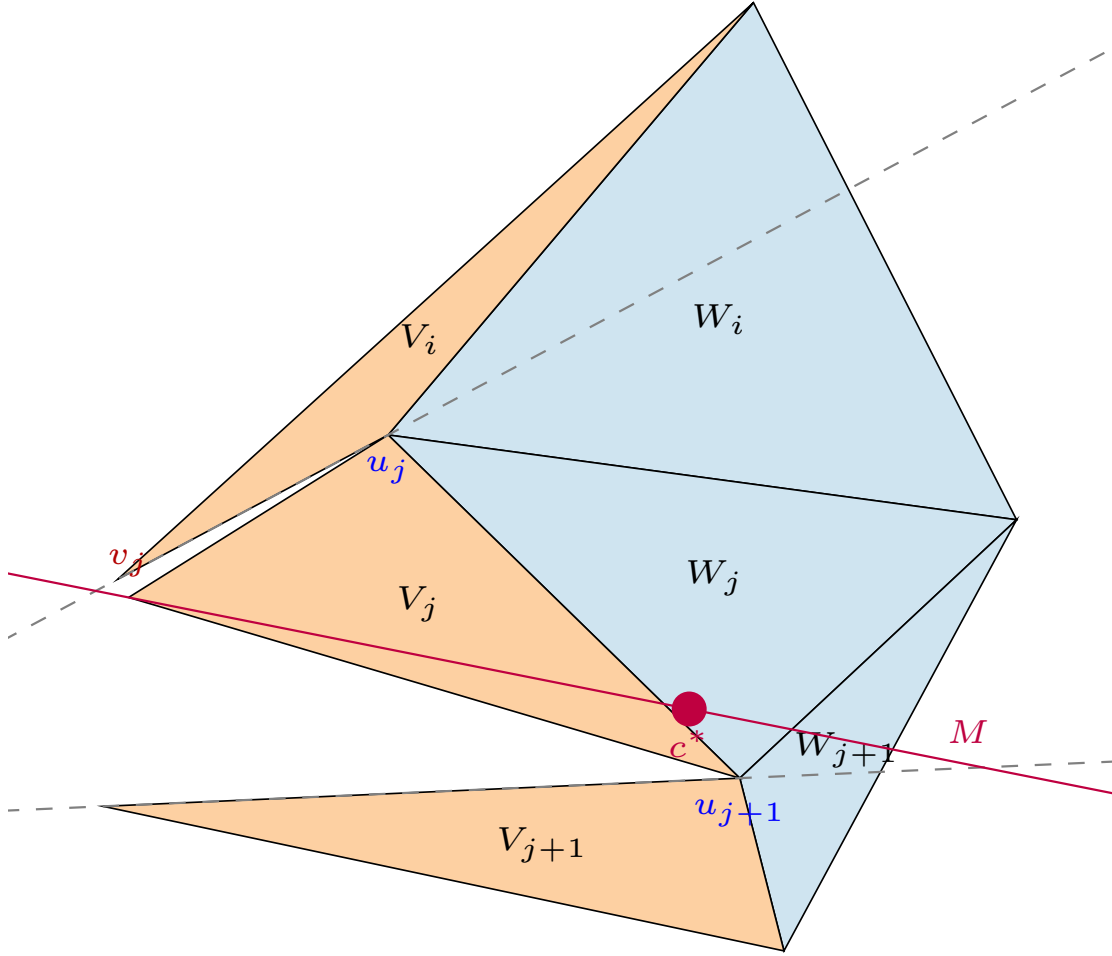


Figure 3: A triple under (H) with $\kappa < \nu_j$: c^* , the line M and the two dashed cut-edge lines; both petals stay on their own side of M .

Superseded: what remained for the rotation argument. Prove that under (H) and $\kappa < \nu_j$ neither $V_i^{\text{fan}} \cap \{\text{above}\}$ nor $V_{j+1}^{\text{fan}} \cap \{\text{above}\}$ crosses M (or, more weakly, that (3) holds). The three crossing mechanisms are: (a) the wedge at v : $\nu_i > \pi - \mu_u$; (b) the far vertex u_i above the base on u' 's side of M ; (c) an edge of V_i crossing the base line outside the segment uu' on u' 's side of M . Mechanism (c) through vu_i requires $\nu_i + \nu_j > \pi + \psi$; (a) requires $\nu_i + \nu_j > \pi$; by Lemma 11 these need $\kappa_v < \pi$ and two of the four angles at v close to $4\pi/7$. The numerics say the fan faces then obstruct this, but the argument is not yet written.

4 Case A is closed: apex cones

The side conditions of §3 are not needed. Two lemmas finish the above-base case.

Lemma 13 (no wrap). *Under (H), $\nu_i + \nu_j \leq \pi + \kappa_u + \kappa_{u'}$ and $\nu_{j+1} + \nu_j \leq \pi + \kappa_u + \kappa_{u'}$ for every triple.*

Proof. From $\sum_x \kappa_x = 4\pi$ and (H), $\kappa := \kappa_u + \kappa_{u'} = 4\pi - \kappa_v - \kappa_w - \kappa_{u_i} - \kappa_{u_m} \geq 4\pi - 4\kappa_v = 4\sum \nu - 4\pi$. If $\sum \nu \geq \pi$ then $\pi + \kappa \geq 4\sum \nu - 3\pi \geq \sum \nu \geq \nu_i + \nu_j$; if $\sum \nu < \pi$ then $\nu_i + \nu_j < \pi < \pi + \kappa$. $\square$

Lemma 14 (apex cones). *Let C_i be the closed cone at v_i spanned by the cut edges $v_i u$ and $v_i u_i$ (it contains V_i), and C_{j+1} the cone at v_{j+1} spanned by $v_{j+1} u'$ and $v_{j+1} u'_m$. If $\kappa < \nu_j$, $\nu_i \leq \pi - \nu_j + \kappa$ and $\nu_{j+1} \leq \pi - \nu_j + \kappa$, then C_i and C_{j+1} have disjoint interiors. In particular V_i and V_{j+1} do not overlap.*

Proof. Measure directions counter-clockwise from $u \rightarrow u'$ and put $\alpha = \phi + \kappa_u$, $\beta = \psi + \kappa_{u'}$, so $\alpha + \beta = \pi - \nu_j + \kappa < \pi$. The cut edge uv_i has direction α (the gap turns v_j away from V_j) and V_i lies on its left, so from v_i the cone C_i is the sector $I_i = [\pi + \alpha - \nu_i, \pi + \alpha]$ ending at the direction e_1 towards u . Likewise $u'v_{j+1}$ has direction $\pi - \beta$, V_{j+1} lies on its right, and C_{j+1} is the sector $I_{j+1} = [-\beta, -\beta + \nu_{j+1}]$ starting at the direction towards u' . Two cones meet iff $d := v_{j+1} - v_i$ lies in the convex cone K generated by $I_i \cup (-I_{j+1})$, where $-I_{j+1} = [\pi - \beta, \pi - \beta + \nu_{j+1}]$. The two hypotheses say exactly that I_i and $-I_{j+1}$ lie inside $S = [\pi - \beta, \pi + \alpha]$, a sector of width $\alpha + \beta < \pi$ bounded by e_1 and by $f_2 =$ the direction of $u' \rightarrow v_{j+1}$; hence $K = S$. Now $d = s_u e_1 + \ell e_0 + s_{u'} f_2$ with e_0 the base direction, and writing $e_0 = c_1 e_1 + c_2 f_2$ gives $c_1 = -\sin \beta / \sin(\alpha + \beta)$, $c_2 = -\sin \alpha / \sin(\alpha + \beta)$. Thus $d = (s_u - |up^*|)e_1 + (s_{u'} - |u'p^*|)f_2$ where p^* is the crossing of the two cut-edge lines, and $d \in S$ iff $s_u \geq |up^*|$ and $s_{u'} \geq |u'p^*|$, i.e. iff both cut edges reach p^* , i.e. iff they cross. Lemma 7 excludes this (equality means the cones touch at p^*). $\square$

Proposition 15. *Under (H) the Triple Lemma holds whenever $\kappa_u + \kappa_{u'} < \nu_j$ (Case A).*

Proof. Lemma 13 gives the two angle conditions of Lemma 14. $\square$

Numerical evidence 16. On 3241 random Case-A triples without hypothesis the cones never meet when the two angle conditions hold, and the conditions fail in 8% of them; under (H) they never fail (208 triples; adversarial margin 1.3 rad, `adv_wrap.py`), and the cone intersection has empty interior in every Case-A triple (adversarial depth -0.0007 diameters at the curvature guard, `adv_cones.py`). Without hypothesis the adversary makes the cones meet (depth $+0.006$), as it must, since 26 above-base overlaps exist.

Case B in this language. Any meeting lies in $C_i \cap C_{j+1}$, and in Case B this intersection can be non-empty, so the cones alone do not suffice; the next section gives the exact structure of Case B.

5 The below-base case (Case B)

Throughout this historical reduction, assume $\kappa = \kappa_u + \kappa_{u'} \geq \nu_j$ and the restricted angle hypotheses of Lemma 5. Only then does that lemma put a meeting on w 's side of the base. The complementary large-angle regimes remain unhandled here. Write $u_k = u_{j+2}$ for the far vertex of W_{j+1} (in Z_k this is the slit copy, $k = i - 1$), A for the line through the outer edge uu_i of W_i , C for the line through $u'u_k$, $\alpha = \angle_u W_i + \angle_u W_j$, $\beta = \angle_{u'} W_j + \angle_{u'} W_{j+1}$ and $\Sigma_W = \alpha + \beta$ (the four fan angles at u, u'). Let $\kappa_{\text{back}} = \kappa_{u_i} + \kappa_{u_k} + \kappa_w$ be the total gap on the slit side, so that $\kappa_v + \kappa + \kappa_{\text{back}} = 4\pi$, and let

$V_m = V_{i-1}$ be the fourth petal, glued to $W_m = W_k$. Within the restricted range above, if $\alpha \geq \pi$ then V_i lies above the base (its angular interval at u ends at $2\pi - \alpha \leq \pi$), so it cannot have a Case-B meeting. The mirror argument applies to $\beta \geq \pi$. In the remaining wedge analysis assume $0 < \alpha, \beta < \pi$.

Lemma 17 (the wedge Ω). *V_i lies in the closed half-plane H_A of A not containing w , and V_{j+1} in the half-plane H_C of C not containing w ; every common point lies in $\Omega := H_A \cap H_C$. If $\Sigma_W \geq \pi$ then Ω lies strictly above the base line, so $\Omega \cap \mathcal{D} = \emptyset$ and there is no Case-B meeting. If $\Sigma_W < \pi$ the lines cross at a point X on w 's side of the base, w lies in the triangle $uu'X$, and Ω is the wedge at X opposite to that triangle, of angle $\theta_X = \pi - \Sigma_W$.*

Proof. V_i and W_i lie on opposite sides of their common edge line A . On the base line, H_A is the ray beyond u away from u' and H_C the ray beyond u' away from u , so Ω misses the base line and lies on one side of it. The rays $u \rightarrow u_i$ and $u' \rightarrow u_k$ make the angles α, β with the base on w 's side; they converge there iff $\Sigma_W < \pi$, and then X and Ω are below the base; otherwise Ω is above (or empty), while $\mathcal{D}$ is below in Case B. w is on u' 's side of A , on u 's side of C and below the base. $\square$

Lemma 18 (how a petal reaches Ω). *Let $\Sigma_W < \pi$. V_i meets the interior of Ω iff u_i lies past X on A (far vertex, f) or $v_i \in \text{int } \Omega$ (apex, a); likewise V_{j+1} with u_k and v_{j+1} . The far vertices u_i, u_k are never both past X .*

Proof. $V_i \subset H_A$ meets the open half-plane of C iff one of its vertices does; u is on w 's side of C , and $u_i \in A$ is beyond C iff it is past X . If both far vertices were past X , then $X \in uu_i \cap u'u_k \subset W_i \cap W_{j+1} = \{w\}$, but $w \notin A$. $\square$

So a meeting needs $\Sigma_W < \pi$ and one of the sub-cases (a, a), (fa, a), (a, fa), (f, a), (a, f): at least one apex lies in Ω .

Lemma 19 (direction condition). *For every triple,*

$$\angle_{u_k} V_{j+1} - \Sigma_W = \nu_i + \nu_m + \angle_u V_i - \kappa_{\text{back}}, \quad \angle_{u_i} V_i - \Sigma_W = \nu_{j+1} + \nu_m + \angle_{u'} V_{j+1} - \kappa_{\text{back}}.$$

If $\Sigma_W < \pi$, u_k is strictly before X , and v_{j+1} is in the interior of Ω , then $\angle_{u_k} V_{j+1} > \Sigma_W$. The mirror statement requires u_i strictly before X . No such direction conclusion is asserted here when the respective far vertex is already past X .

Proof. $\alpha = 2\pi - \kappa_u - \phi - \angle_u V_i$, $\beta = 2\pi - \kappa_{u'} - \psi - \angle_{u'} V_{j+1}$, $\phi + \psi = \pi - \nu_j$ and $\angle_{u_k} V_{j+1} = \pi - \angle_{u'} V_{j+1} - \nu_{j+1}$ give $\angle_{u_k} V_{j+1} - \Sigma_W = \kappa + \angle_u V_i - \nu_j - \nu_{j+1} - 2\pi$, and $\kappa = 2\pi + \sum \nu - \kappa_{\text{back}}$. For the direction: $u_k \in C$ lies on w 's side of A and $v_{j+1} \in \Omega$, so the direction $u_k \rightarrow v_{j+1}$ points to the far side of C (the open half-circle of directions on V_{j+1} 's side of C) and across A ; the direction of A towards u_i makes the angle Σ_W with $u_k \rightarrow u'$ (exterior angle at X), so the admissible directions are those making an angle in (Σ_W, π) with $u_k \rightarrow u'$, and $u_k \rightarrow v_{j+1}$ makes the angle $\angle_{u_k} V_{j+1}$. $\square$

Lemma 20 (sub-case (a,a): the hexagon identity). *Let $\Sigma_W < \pi$, neither far vertex past X , $v_i, v_{j+1} \in \Omega$, and assume the three local pairs at the slit do not overlap (Lemma L). If p is a common point of V_i and V_{j+1} , then*

$$\nu_m = \kappa_{\text{back}} + \gamma_p + \angle(p u_i v_i) + \angle(p u_k v_{j+1}), \quad \gamma_p > 0,$$

where γ_p is the angle at p of the hexagon $u_i p u_k w u'_k v_m$. In particular a meeting needs $\nu_m > \kappa_{\text{back}}$.

Proof. The closed curve $\Gamma = u_i \rightarrow p \rightarrow u_k \rightarrow w \rightarrow u_i$ (segments in $V_i, V_{j+1}, W_{j+1}, W_i$) is simple: $[w, u_k]$ lies on w 's side of A and meets $V_i \subset H_A$ at most at X , symmetrically for $[w, u_i]$. It lies in the triangle $\Delta = uu'p$, which is on w 's side of the base line because $p \in \Omega$. Let Q be the region bounded by Γ . Its angle at w is $\omega_m + \kappa_w$ or $\omega_i + \omega_j + \omega_{j+1}$; in the second case W_j , hence V_j (which meets Γ nowhere, by the shared-vertex lemma), lie in $Q \subset \Delta$, but v_j is above the base. So Q contains W_m near w , hence W_m and V_m (their boundaries do not cross Γ : shared vertices with V_i, W_i, W_{j+1} , and Lemma L for the pairs with V_{j+1}, W_{j+1}). The hexagon $R = Q \setminus (W_m \cup V_m)$ has interior angles $\angle_{u_i} V_i - \angle(p u_i u) + \kappa_{u_i}$ at u_i , γ_p at p , $2\pi - \angle_{u_k} W_{j+1} - \angle(p u_k u')$ at u_k , κ_w at w , $2\pi - \angle_{u'_k} W_m - \angle_{u'_k} V_m$ at u'_k and $2\pi - \nu_m$ at v_m ; their sum is 4π , and $2\pi - \angle_{u_k} W_{j+1} - \angle_{u'_k} W_m - \angle_{u'_k} V_m = \kappa_{u_k} + \angle_{u_k} V_{j+1}$. $\square$

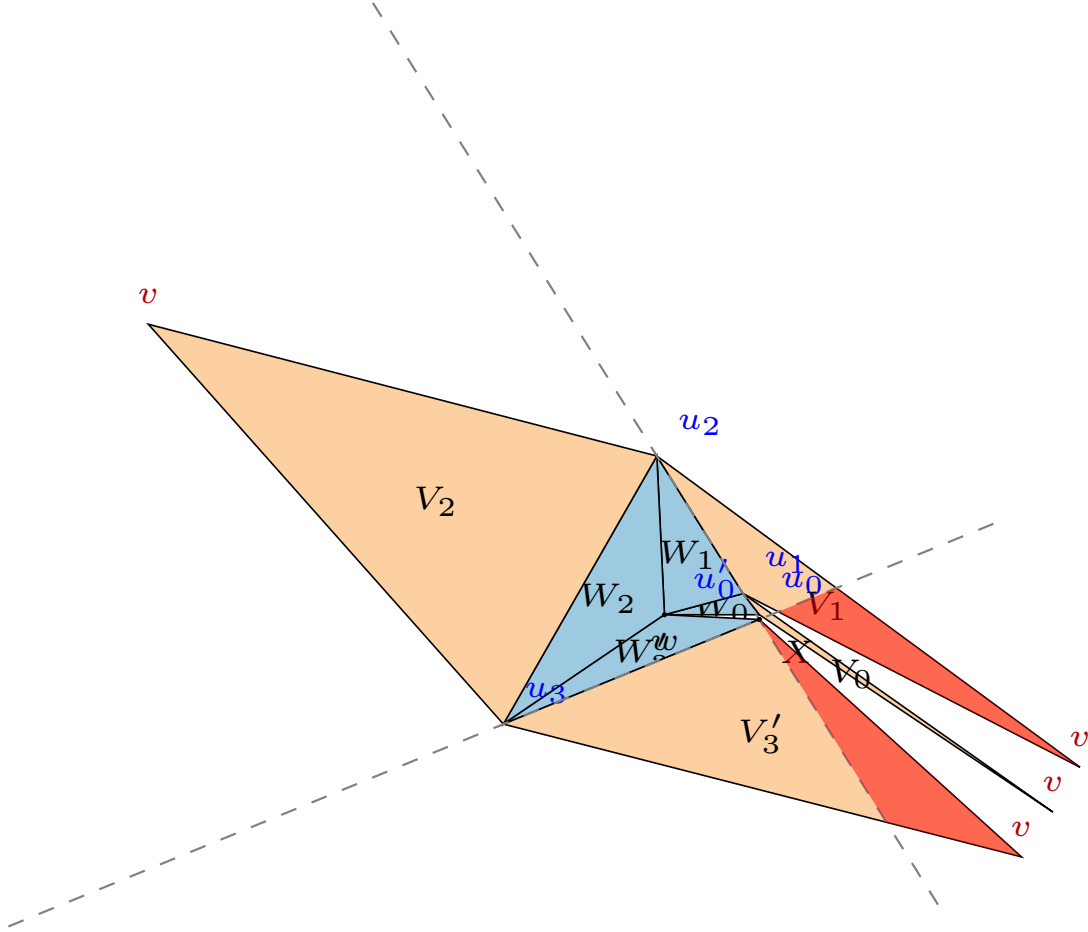


Figure 4: The tightest sub-case (a,a) configuration under (H) (`caseB_sub.py`): w, u_i, u_k nearly flat, u, u' needles, $\Sigma_W = 1.74$, the far vertex u_k at X . The two petal parts inside Ω (red) are separated by the thin fourth petal V_m ; closest approach 3.5% of the diameter.

Regime split for (a,a). If $\nu_m \leq \kappa_{\text{back}}$ Lemma 20 excludes a meeting. If $\nu_m > \kappa_{\text{back}}$, the two outer cut edges $u_i v_i$ and $u_k v_{j+1}$ make the angles $\angle_{u_i} V_m + \kappa_{u_i}$ and $\angle_{u'_k} V_m + \kappa_{u_k} + \kappa_w$ with the segment $u_i u_k$, so their lines converge on V_m 's side with angle $\nu_m - \kappa_{\text{back}}$, and Lemma 14 applies verbatim with the base uu' replaced by $u_i u_k$ (no-wrap $\nu_i, \nu_{j+1} \leq \pi - \nu_m + \kappa_{\text{back}}$ follows from (H) as in Lemma 13): the cones C_i, C_{j+1} have disjoint interiors unless the two outer cut edges cross.

Two gaps remain here: (i) a no-cross lemma for the outer cut edges across the slit ($u_k \neq u'_k$; the argument of Lemma 7 needs $|u_k v_m| \geq |u'_k v_m|$); (ii) the sub-cases with a far vertex past X .

Numerical evidence 21. Under (H), in 23,578 Case-B triples, 19,858 have $\Sigma_W \geq \pi$ and only 27 have both petals reaching Ω (sub-cases (a,a) 22, (fa,a) 4, (a,fa) 1); no (f,f), as it must be. Seeded adversarial closest approach inside Ω (`caseB_sub.py adv`): (a,a) 3.5%, (fa,a) 3.2%, (a,fa) 1.9% of the diameter (the last slightly past the guard). Without hypothesis (a,fa) overlaps (+1.6%) and (a,a) stays 0.7% apart. In every (a,a) triple with $\nu_m > \kappa_{\text{back}}$ (109 found, with and without (H)) the apex cones have disjoint interiors, and the outer cut edges never cross when $\nu_m > \kappa_{\text{back}}$ (adversarial best -0.003 diameters, `adv_crossA.py`); $\nu_m > \kappa_{\text{back}}$ does occur under (H) in (a,a) (excess up to 0.30 rad), so both regimes are needed. An earlier count “15% of Case-B triples have both petals reaching the cone intersection” used a vacuous test for the position of X and is withdrawn.

Historical numerical evidence for the slit rule. The old `adv_angle.py` mode `i` used the wrong base vertex; its saved margins are withdrawn for the intended far-angle claim. The code is corrected, but the original searches have not all been rerun. The direct half-plane probes below are different experiments. Lemma F is used with the slit at the sharpest neighbour u_k of w ($\kappa_{u_k} \geq \kappa_{u_t}$ for all t ; Lemma L needs this rule: without it the local pairs overlap under (H) in 3.4% of nets). Under (H) and the rule, in 641 Case-B triples with $\Sigma_W < \pi$ the petal V_i never reaches Ω and V_{j+1} reaches it only through its far vertex; adversarially (`adv_reach.py`, `adv_angle.py`): v_i stays 0.35 diameters short of C (with $\Sigma_W \leq \pi - 0.3$), v_{j+1} stays 0.12 diameters short of A , u_i stays 0.026 short of X , and under the rule alone $\angle_{u_k} V_{j+1} \leq \Sigma_W - 0.35$. By Lemma 18 it therefore suffices to prove, under (H) and the rule, that neither apex lies in Ω : **(P1)** $v_{j+1} \notin \Omega$, **(P2)** $v_i \notin \Omega$. Linear programming over the 24 face angles (face sums, cone inequalities, curvature bounds, the rule and (H)) numerically allows $\angle_{u_k} V_{j+1} - \Sigma_W$ up to $\pi/2$. This suggests that these linear constraints alone are insufficient; additional metric or angle-realizability constraints may be needed. The tight sampled configurations are needle-shaped (Figure above). This is the recommended route: it replaces the five sub-cases by two apex statements within the stated angle regime. It does not settle the large-angle regimes or Lemma L.

6 Known false / do not retry

Simple separating lines for the two petals inside $\mathcal{D}$ (bisector of $\mathcal{D}$ at p^* , lines through v_j) fail in about a third of the non-degenerate cases even under (H): the separation is a rotation phenomenon, not a half-plane one. The six wedge-lean inequalities of the handoff fail in 0.5% of nets under (H) (needle-like w), so they cannot replace the argument above.

Code

`octa.py` (module: all quantities, nets Z_k , nine pairs, intersection polygons), `vwedge.py`, `oppcases.py` (collect/classify/draw overlaps), `adv_opp.py` (adversarial separation), `cstar.py` (checks of §3), `adv_eps.py`, `adv_mu.py`, `adv_cstar.py` (adversarial tests of (3) and of the crossing conditions), `tikz_net.py` (figures), `draw.py` (PNG); Case B: `caseB_sub.py` (sub-cases, seeded adversary), `caseB_draw.py`, `adv_crossA.py`, `adv_reach.py`, `adv_angle.py`, `adv_rule.py`, `lp_angles.py`.